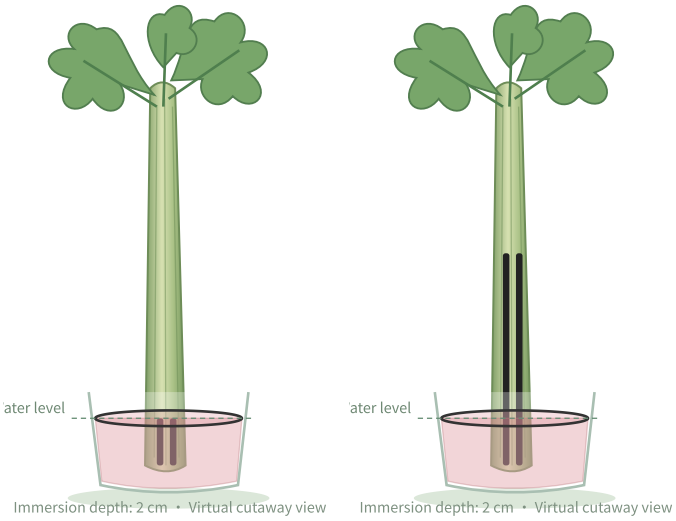
Name: _____ Class and student number: _____

The red water trace in celery

What the school gardening group found



After some time

Celery leaf stalk (petiole (葉柄))

This part looks like a stem. It contains xylem (木質部), so we can observe water moving upwards. It is a leaf stalk, called a petiole (葉柄).

This diagram shows an imaginary cutaway so you can see the water trace inside. The celery used in the experiment has not been cut lengthwise.

What can the red water trace tell us?

The gardening group puts celery with leaves in red water. They observe it before and after some time has passed.

Look at the two pictures on the left and record the changes you see. The group wants to know: do light and leaves affect water transport?

Main investigation question

When other conditions are the same, how does light intensity (光強度) affect the upward speed of the red water trace in celery with leaves?

Your first observations

What changes do you see? Describe them first. You do not need to explain why yet.

Complete your prediction

If light intensity (光強度) increases, the upward speed of the red water trace will （increase ／ decrease ／ show no clear change）.

Circle one answer in each set of brackets in the sentence.

My reason

Write your ideas. Your prediction does not have to match the expected result.

Identify the variables

Choose the factors that fit each variable type. Save your ideas now. You can view feedback after you submit.

Independent variable (the factor changed on purpose)

- ☐ Light intensity (光強度)
- ☐ Height / average upward speed of the red water trace
- ☐ Temperature
- ☐ Humidity (濕度) and air movement
- ☐ Red water concentration and depth of immersion (浸入深度)
- ☐ Total leaf area of the setups with leaves
- ☐ Length and thickness of the celery stalks

Dependent variable (the result measured)

- ☐ Light intensity (光強度)
- ☐ Height / average upward speed of the red water trace
- ☐ Temperature
- ☐ Humidity (濕度) and air movement
- ☐ Red water concentration and depth of immersion (浸入深度)
- ☐ Total leaf area of the setups with leaves
- ☐ Length and thickness of the celery stalks

Controlled variables (the factors kept the same)

- ☐ Light intensity (光強度)
- ☐ Height / average upward speed of the red water trace
- ☐ Temperature
- ☐ Humidity (濕度) and air movement
- ☐ Red water concentration and depth of immersion (浸入深度)
- ☐ Total leaf area of the setups with leaves
- ☐ Length and thickness of the celery stalks

What assumptions does this investigation make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all the assumptions that apply. You may choose more than one.

- ☐ Changing the lamp distance does not make the setups have different temperatures.
- ☐ Setups A, B and C have the same total leaf area.
- ☐ Leaf area does not affect the results, so it does not need to be kept the same.
- ☐ Humidity (濕度) and air movement are the same for all setups.

The control group

How would you plan a comparison setup to find out whether leaves help with water transport?

(Hint: the control group provides a basis for comparison. It helps check whether the change in the experiment is caused by the independent variable rather than other external factors. The control group therefore does not receive the treatment being tested.)

(Hint: how is the control setup different from the experiment setup?)

Draw and label your four setups

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.

Include the celery, leaves, red water, lamps, distances and measurement positions. Put 2 cm of each celery leaf stalk into the red water.


Celery ×4


Desk lamp ×4


Cup of red water ×4



30 cm ruler ×4


Scissors ×1



Timer ×1

Setup labels / written design

Add labels for your drawing, or describe all four setups in words instead of drawing them.

Save the original design before giving out B. Students keep a copy of A for reflection.